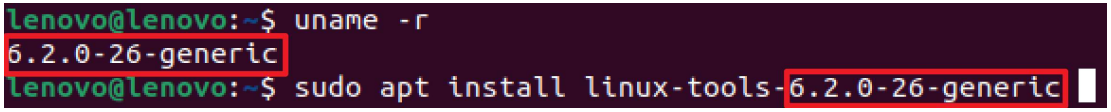
练习4：学习性能分析工具Perf的基本使用

- **任务**：通过 `perf` 观察程序性能。
- **子任务**：
  - 通过 `perf list` 命令查看支持的性能事件；
  - 通过 `perf stat` 查看练习2的矩阵乘法程序的缓存使用情况。
- **目标**：学会 `perf` 的基本使用。

1. 命令安装 `perf`

`perf` 工具包含在 `linux-tools` 中，安装时需保证其版本号与Linux内核版本号一致。

以实验室的Ubuntu双系统为例，安装前执行 `uname -r` 查看内核版本号，再通过 `apt` 命令安装，如下图所示。



实验室Ubuntu双系统的root密码是 `user1234`。

> ⚠ **上图仅为示例，需以命令执行的实际输出为准**
>
> 使用 `apt` 安装 `perf` 时，“`linux-tools-`”后面的版本号信息需以 `uname -r` 的输出为准，请自行作相应的修改。

若提示“`E: Unable to locate package linux-tools-xxx`”，可尝试安装邻近的版本或自行搜索其他解决方案。

2. 查看 `perf` 支持的性能事件

```
sudo perf list
```

若执行失败，可尝试执行 `perf list`。

```
lenovo@10:~$ perf list

List of pre-defined events (to be used in -e):

  duration_time                                      [Tool event]

  branch-instructions OR cpu/branch-instructions/    [Kernel PMU event]
  branch-misses OR cpu/branch-misses/                [Kernel PMU event]
  bus-cycles OR cpu/bus-cycles/                      [Kernel PMU event]
  cache-misses OR cpu/cache-misses/                  [Kernel PMU event]
  cache-references OR cpu/cache-references/          [Kernel PMU event]
  cpu-cycles OR cpu/cpu-cycles/                      [Kernel PMU event]
  cstate_core/c3-residency/                          [Kernel PMU event]
  cstate_core/c6-residency/                          [Kernel PMU event]
  cstate_core/c7-residency/                          [Kernel PMU event]
  cstate_pkg/c10-residency/                          [Kernel PMU event]
  cstate_pkg/c2-residency/                           [Kernel PMU event]
  cstate_pkg/c3-residency/                           [Kernel PMU event]
  cstate_pkg/c6-residency/                           [Kernel PMU event]
  cstate_pkg/c7-residency/                           [Kernel PMU event]
  cstate_pkg/c8-residency/                           [Kernel PMU event]
  cstate_pkg/c9-residency/                           [Kernel PMU event]
  cycles-ct OR cpu/cycles-ct/                        [Kernel PMU event]
  cycles-t OR cpu/cycles-t/                          [Kernel PMU event]
  el-abort OR cpu/el-abort/                          [Kernel PMU event]
  el-capacity OR cpu/el-capacity/                    [Kernel PMU event]
  el-commit OR cpu/el-commit/                        [Kernel PMU event]
  el-conflict OR cpu/el-conflict/                    [Kernel PMU event]
  el-start OR cpu/el-start/                          [Kernel PMU event]
  i915/actual-frequency/                             [Kernel PMU event]
  i915/bcs0-busy/                                    [Kernel PMU event]
  i915/bcs0-sema/                                    [Kernel PMU event]
  i915/bcs0-wait/                                    [Kernel PMU event]
  i915/interrupts/                                   [Kernel PMU event]
  i915/rc6-residency/                                [Kernel PMU event]
  i915/rcs0-busy/                                    [Kernel PMU event]
  i915/rcs0-sema/                                    [Kernel PMU event]
  i915/rcs0-wait/                                    [Kernel PMU event]
  i915/requested-frequency/                          [Kernel PMU event]
  i915/vcs0-busy/                                    [Kernel PMU event]
  i915/vcs0-sema/                                    [Kernel PMU event]
  i915/vcs0-wait/                                    [Kernel PMU event]
  i915/vecs0-busy/                                   [Kernel PMU event]
  i915/vecs0-sema/                                   [Kernel PMU event]
  i915/vecs0-wait/                                   [Kernel PMU event]
  instructions OR cpu/instructions/                  [Kernel PMU event]
  intel_bts//                                        [Kernel PMU event]
  intel_pt//                                         [Kernel PMU event]
```

`perf list` 命令输出的信息较长，每按一下回车就会多输出一行信息。按Q键即可退出。

3. 查看 `lab1_gemm` 程序的缓存使用情况

先通过 `cd` 命令进入 `lab1/build` 文件夹，再执行下列命令查看基本性能事件：

```
sudo perf stat ./dist/bins/lab1_gemm 256 256 256
```

```
lenovo@10:~/test1/TA/try/computer_archtecture_labs/build$ sudo perf stat ./dist/bins/lab1_gemm 256 256 256
[sudo] lenovo 的密码:
GEMM performance info:
                M, K, N: 256, 256, 256
                Ops: 0.0335544
                Total compute time(s): 3.50527
                Cost(s): 0.0175263
                Benchmark(Gflops): 1.91452

 Performance counter stats for './dist/bins/lab1_gemm 256 256 256':

          3,679.90 msec task-clock                #    0.998 CPUs utilized
               280      context-switches          #    0.076 K/sec
                 0      cpu-migrations            #    0.000 K/sec
               313      page-faults               #    0.085 K/sec
    16,377,257,905      cycles                    #    4.450 GHz
    56,534,878,817      instructions              #    3.45  insn per cycle
     3,541,484,913      branches                  #  962.385 M/sec
        13,853,819      branch-misses             #    0.39% of all branches

       3.687359278 seconds time elapsed

       3.680092000 seconds user
       0.000000000 seconds sys
```

添加 `-e` 选项，可查看指定的性能事件。所支持的性能事件可由 `perf list` 查看。

```
sudo perf stat -e L1-dcache-loads,L1-dcache-load-misses,dTLB-loads,dTLB-load-misses ./dist/bins/lab1_gemm 256 256 256
```

```
lenovo@10:~/test1/TA/try/computer_archtecture_labs$ sudo perf stat -e L1-dcache-loads,L1-dcache-load-misses,dTLB-loads,dTLB-load-misses
 ./build/dist/bins/lab1_gemm 256 256 256
GEMM performance info:
               M, K, N: 256, 256, 256
               Ops: 0.0335544
               Total compute time(s): 3.50997
               Cost(s): 0.0175498
               Benchmark(Gflops): 1.91195

 Performance counter stats for './build/dist/bins/lab1_gemm 256 256 256':

     7,065,850,361      L1-dcache-loads
       232,187,035      L1-dcache-load-misses     #    3.29% of all L1-dcache hits
     7,065,850,361      dTLB-loads
             1,249      dTLB-load-misses          #    0.00% of all dTLB cache hits

       3.689693733 seconds time elapsed

       3.689569000 seconds user
       0.000000000 seconds sys
```